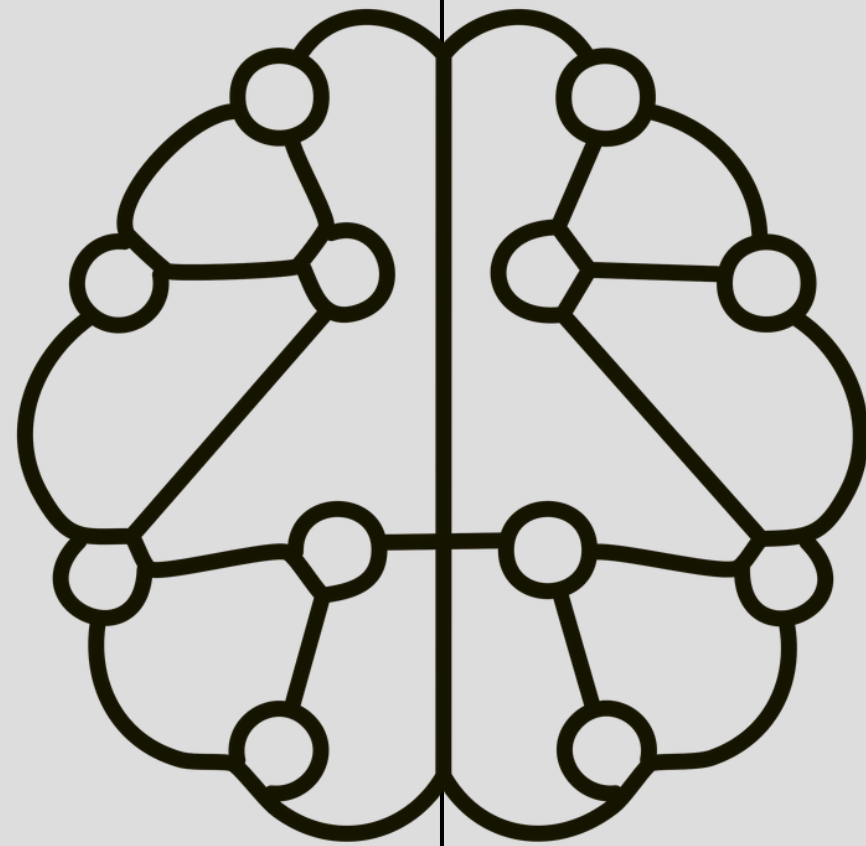
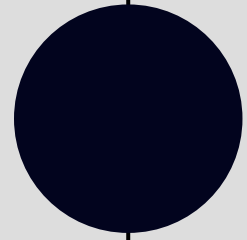
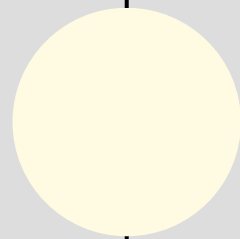




AI Chatbot for reducing teen stress, financial barriers, and academic disengagement in college enrollment.

Bruno Diaz Morales and Ajay Goverdhan



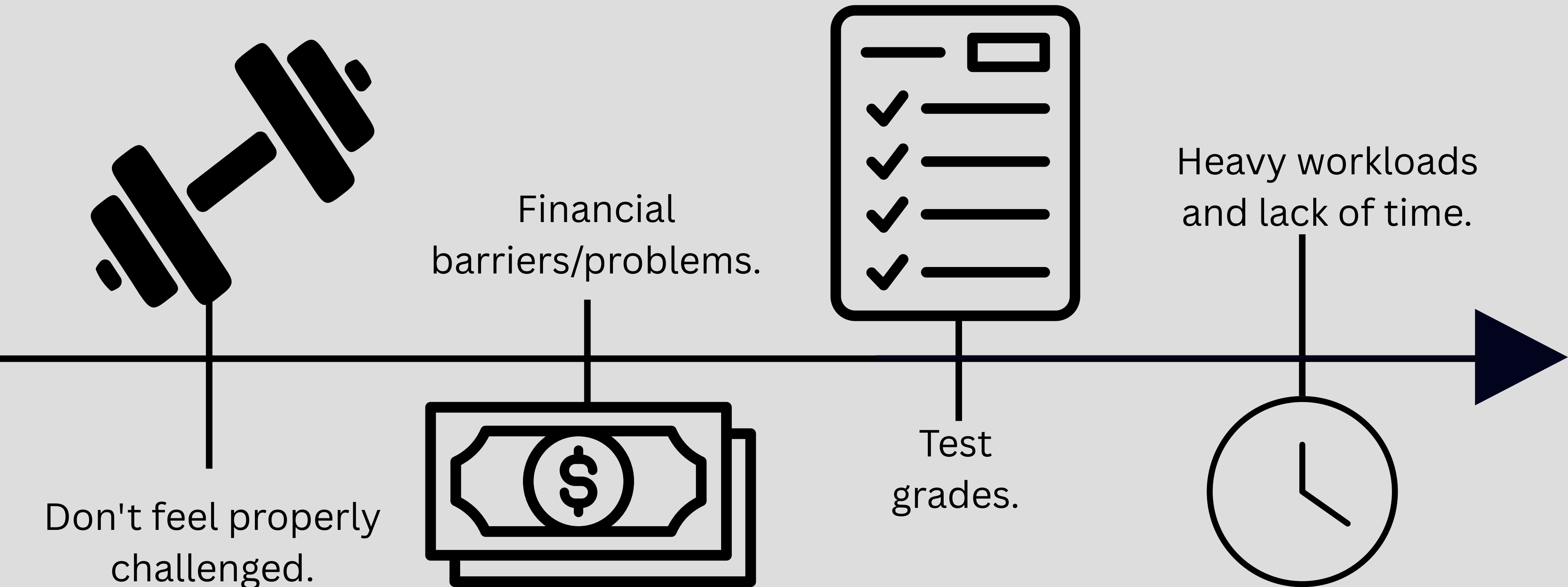
**DESIGNING
THE FUTURE**



Frontiers I
Using ChatGPT API &
Python Programming for
Conversational AI

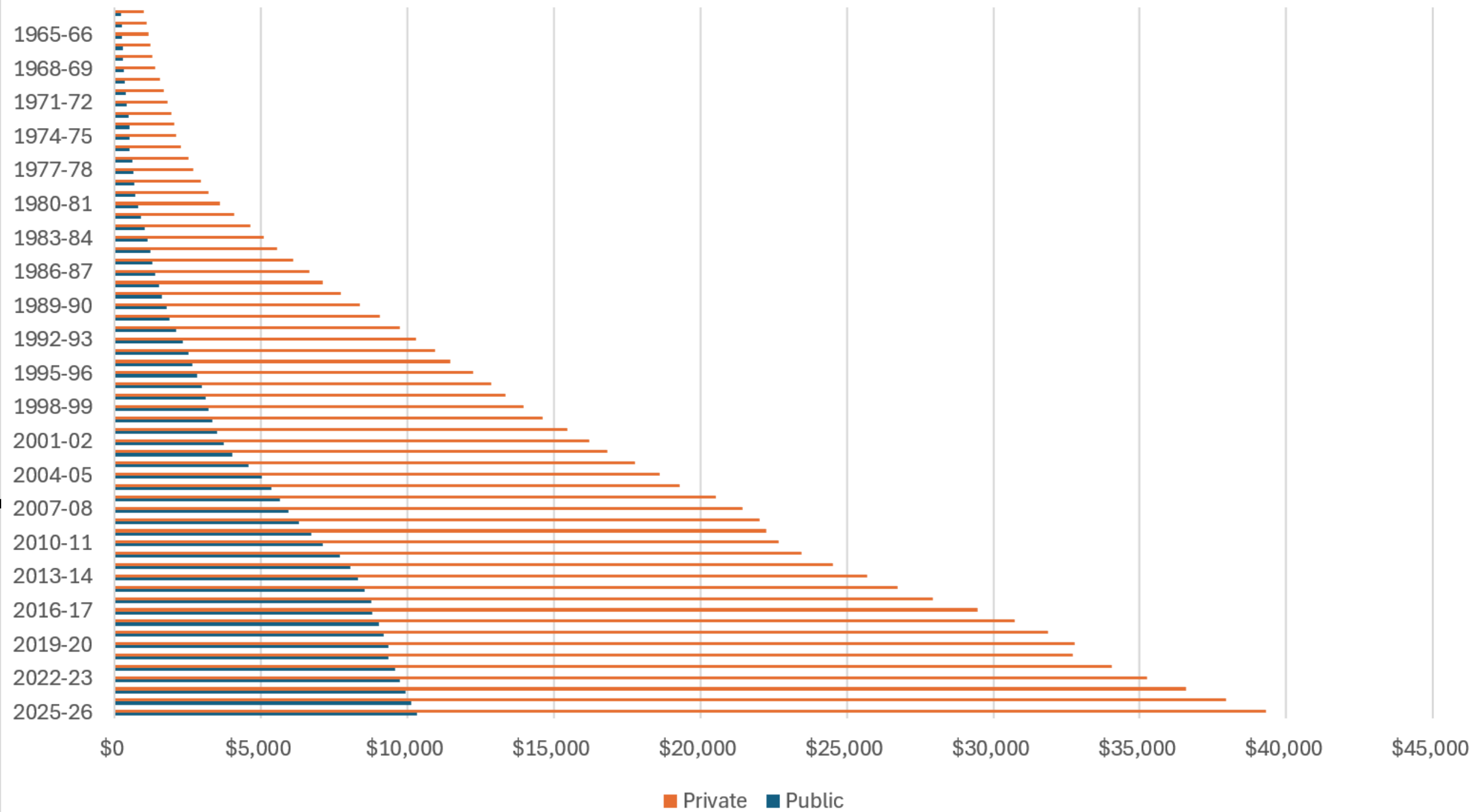
Motivational Background

In recent years, many students have had less interest in college.



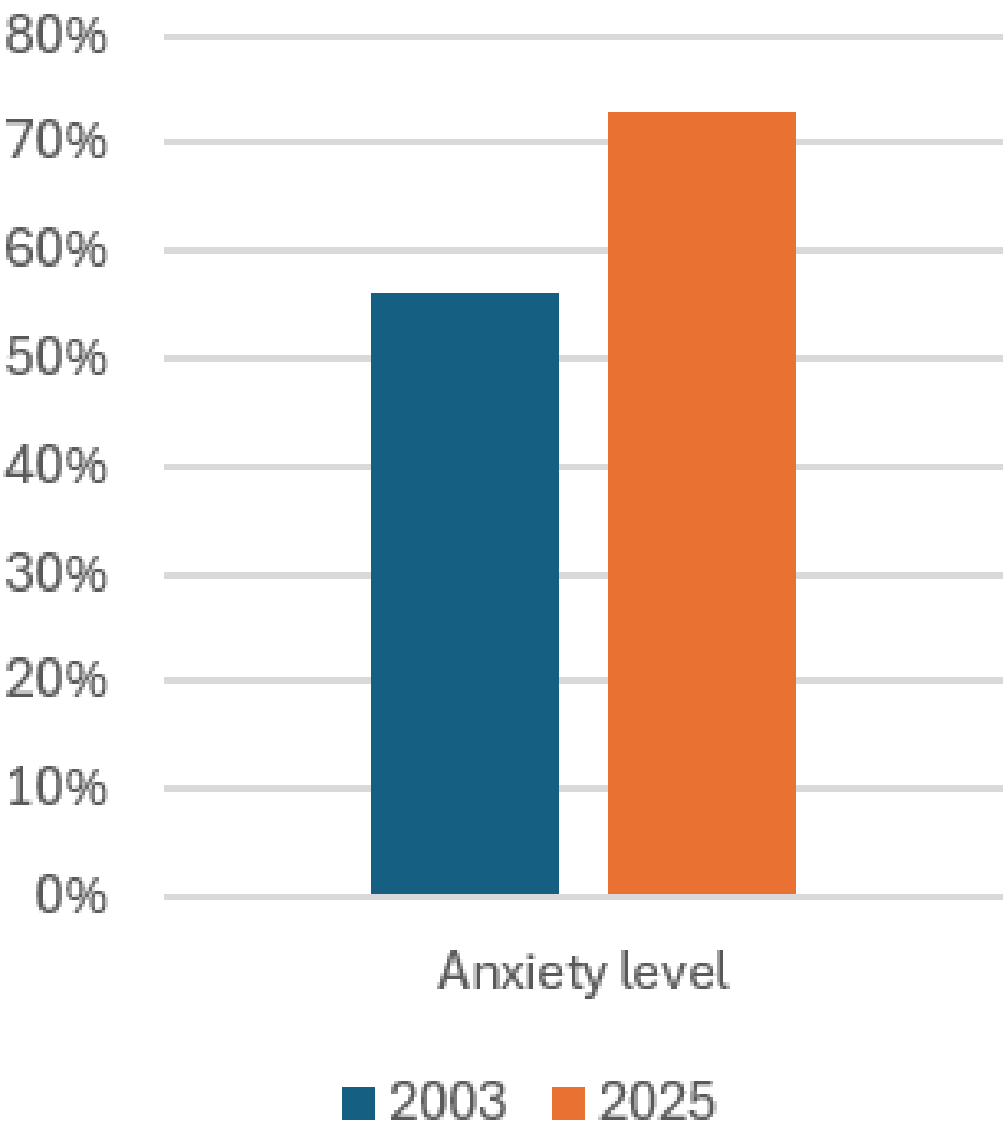
Proven Data

College prices over the years



Source: EducationData.org

Anxiety level while doing college applications



Source: www.wgu.edu

AI-Powered TFA Chatbot

- Chatbot that addresses the needs of students.



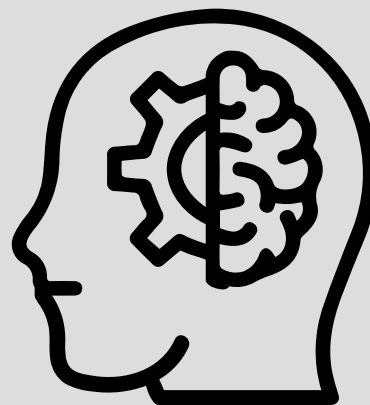
- College process.



- HELP, not do.



- Image generation fully implemented.



Welcome to the College Counseling Chatbot

hello

Hello! How can I assist you today with your college admissions questions?

how can I get into college

Getting into college involves several key steps. Here's a general guide to help you navigate the process:

1. **Research Schools:** Identify colleges that fit your interests, academic goals, and career aspirations. Consider factors such as location, size, programs offered, and campus culture.
2. **Meet Admission Requirements:** Check the admission requirements for the schools you're interested in. This typically includes high school transcripts, standardized test scores (like the SAT or ACT), letters of recommendation, and personal essays.

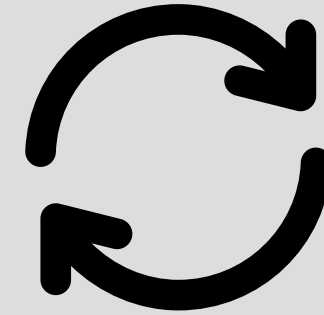
Ask any college-related question



Goals



Used alongside
other resources



Not designed to
replace.

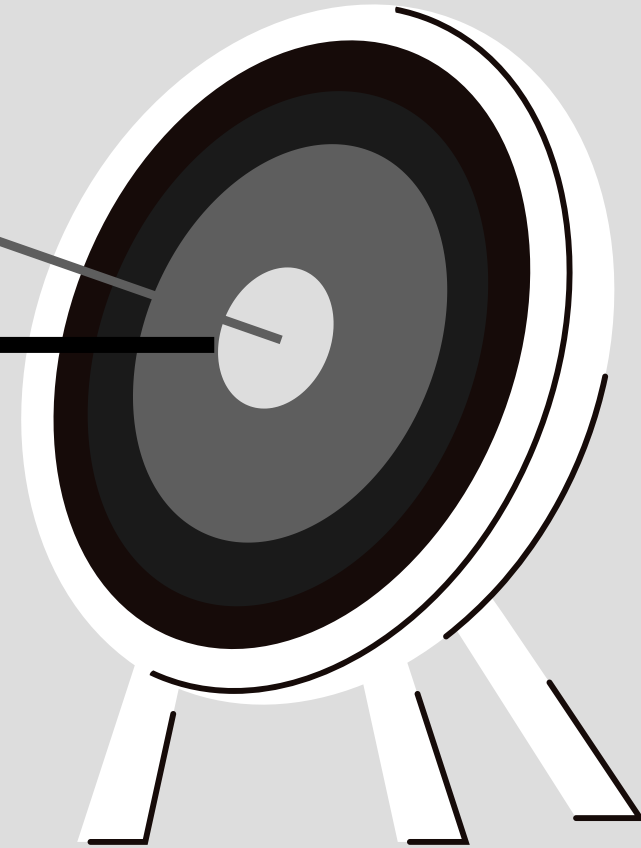
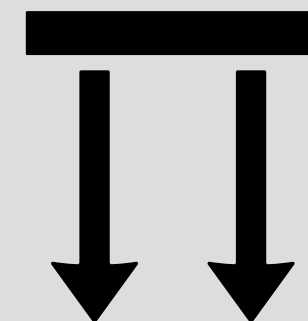
Increase
engagement/
motivation.



Use it as
a tool.



Reduce
application-
related stress.



Model Structure

User inputs question

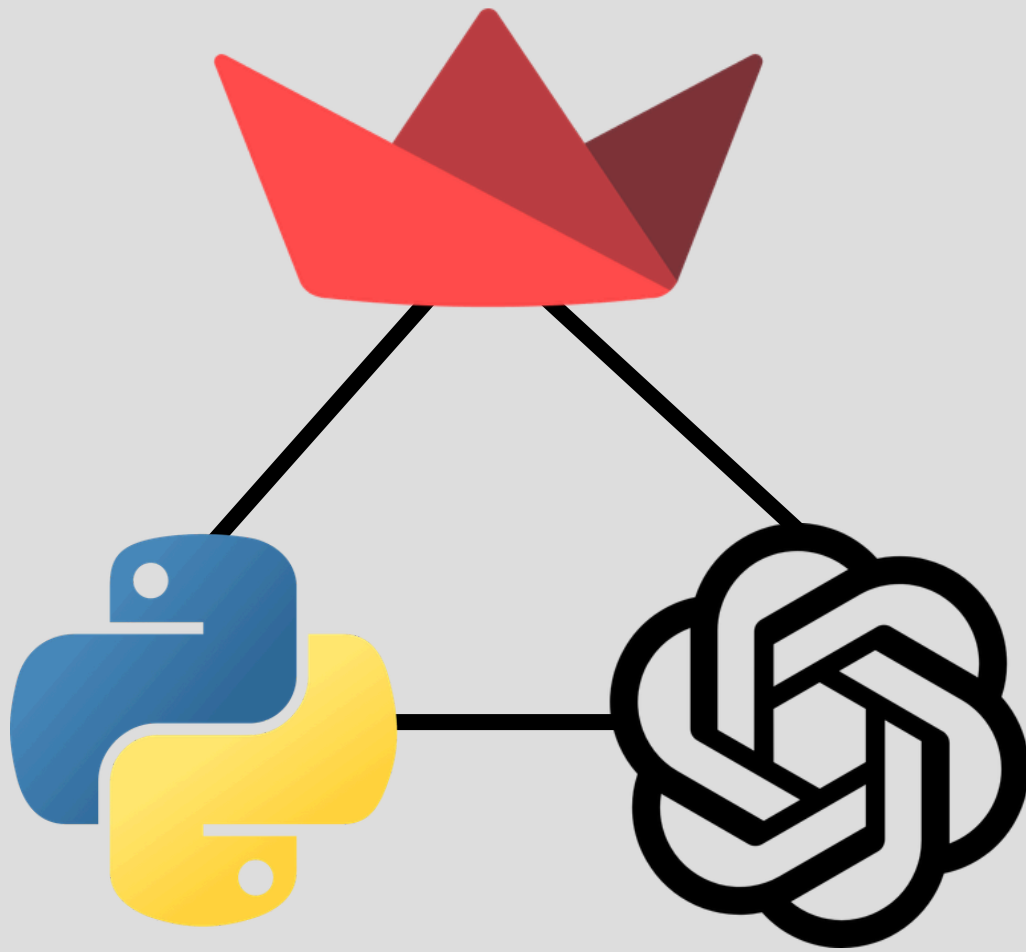
Model reads .json file with the user's information.

Model applies restrictions and customizes answer.

Decides whether it should give text or image as an answer.

Update the UI to display a visually appealing response.

Provides answer to the user.



Tools used to develop this project.

Questions

Text-related questions

Enhanced Prompt: I'm a rising senior who is currently working on the college admissions process. First, try to analyze my profile.. Based on the contents provided in my profile, please suggest at least 5 good extracurriculars I could take to help with college in a clear and supportive way. Then, provide some links to each extracurricular so I can look further into them. Make sure that the response is clear, formatted, and supportive. Also, don't directly tell me what to choose, you can only suggest and explain, but I make the call to what to do at the end of the day. A good example would be providing 5 extracurriculars near my area, summarizing each one in a clear, informative, and supportive way, and then providing links to all 5 of them once you're done summarizing everything.

Expected Answer: Judging by the resume provided, I would recommend you take:

- 1)_____
- 2)_____
- 3)_____

Based on your profile, here are five extracurricular activities that can enhance your college application, especially considering your interests in Computer Science and current involvement in robotics and sports. These suggestions can help you develop both your technical skills and leadership experience, which are valuable for college admissions.

1. Robotics Competitions

...

Link to explore: FIRST Robotics

2. Computer Science Clubs or Coding Meetups

...

Link to explore: Code.org

3. Tech Workshops or Summer Camps

...

Link to explore: Summer Camp Directory

4. Volunteer for STEM Education Programs

...

Link to explore: VolunteerMatch

5. Online Hackathons or Competitions

...

Link to explore: Devpost

Next Steps:

Questions

Text-related questions

Enhanced Prompt: I am a rising-senior student who is looking for new college opportunities that align with what I like and am passionate about. Answer whether (college) offers any special programs or accommodations for students. Include information about academic support programs, disability or accessibility accommodations, tutoring, mentoring, honors programs, first-generation student support, or other student services if available. Explain how students can contact the correct office, such as accessibility services, student affairs, or admissions, to confirm details. If the information may vary by campus, program, or student need, make that clear. Avoid vague answers like “they probably do.” Instead, provide a helpful and transparent response that encourages checking the official college website or contacting the college directly.

Expected Answer: Yes, (college) offer special programs and accommodations. These may include disability/accessibility accommodations, tutoring, academic advising, honors programs, mentoring, or first-generation student support. For the most accurate details, check college’s Accessibility Services, Student Support, or Admissions office on the official website.

Worcester Polytechnic Institute (WPI) offers a range of programs and services aimed at supporting student success across various needs and interests. Here's an overview of the support services you might find helpful:

1. **Academic Support Programs:** WPI offers resources such as academic advising, study skills workshops, and supplemental instruction sessions to help students excel in their courses. The Academic Success Center is a key resource that provides guidance for navigating academic challenges.
2. **Disability and Accessibility Accommodations:** WPI has an Office of Accessibility Services that works to ensure that students with disabilities have equal access to all programs and services. This office coordinates accommodations such as extended testing time, assistive technology, and physical accessibility options. Students can reach out to the office to discuss their specific needs and the process for requesting accommodations.
3. **Tutoring and Mentoring:** WPI provides tutoring services for various subjects. Students can also take advantage of mentoring programs, where peer mentors and faculty sponsors can provide guidance and advice regarding academic and professional development.

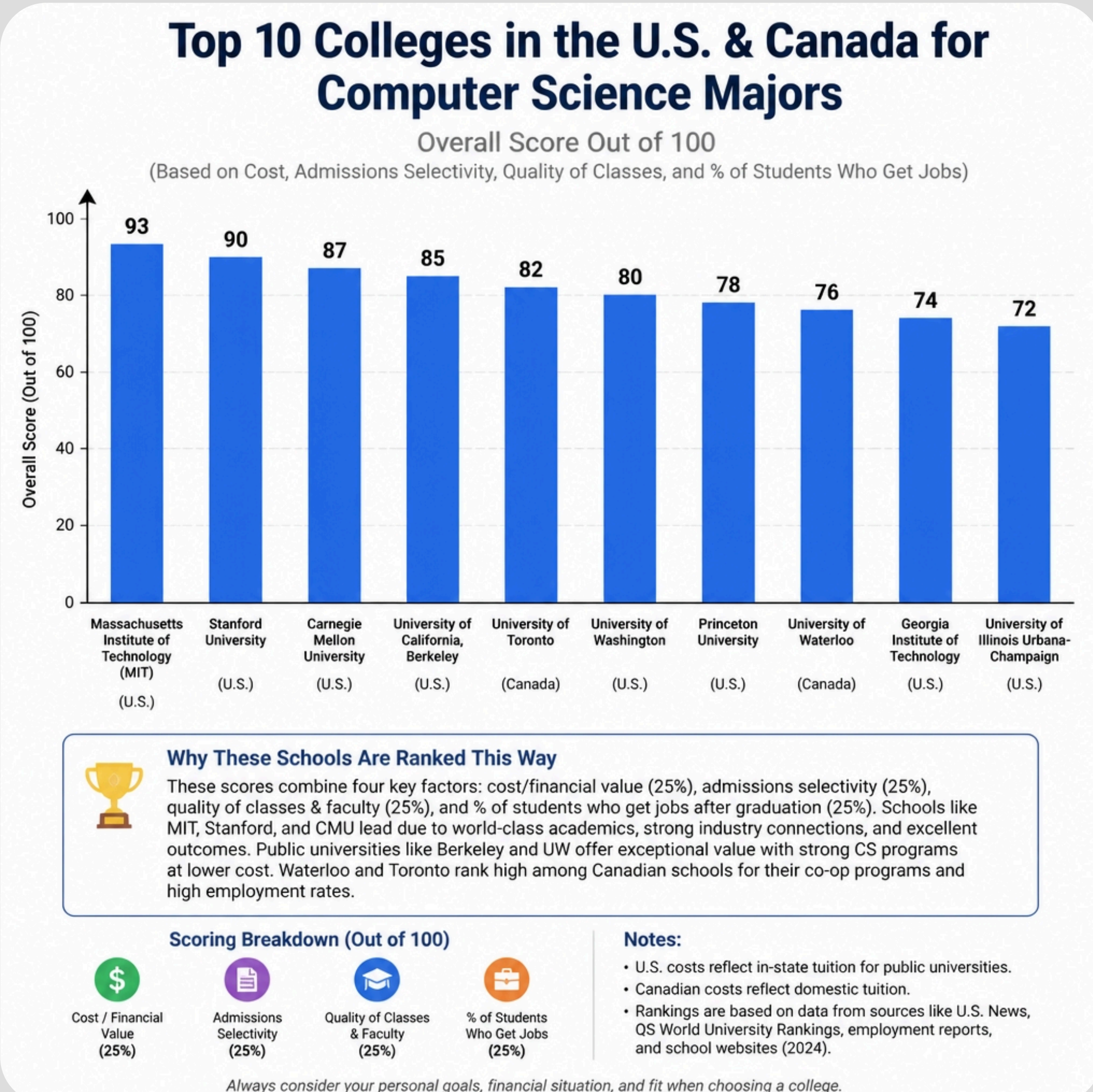
.....

Questions

Image-related questions

Enhanced Prompt: I am a rising senior who is trying to apply to college for a CS major. Please generate a bar graph image for me showing which colleges in the US and Canada are best for CS students. Please take into consideration the costs, requirements to get in, the quality of classes, and % of students who get jobs. Also, make sure to keep the graph easy to read and understandable to anyone. If you can, try to explain why you ranked certain colleges the way you did. A good graph would be a simple bar graph with the top ten best cs schools, with the bottom being lower and the top being better

Expected Answer: A graph that ranks CS majors based on what AI courses they offer



Questions

Image-related questions

Enhanced Prompt: Create a clean, modern comparison infographic image showing a high school student evaluating two colleges: Worcester Polytechnic Institute (WPI) and Georgia Tech. The image should visually compare both schools for a student interested in Computer Science, robotics, engineering, scholarships, affordability, and overall fit. Use a split-screen layout:

- Left side: WPI, represented by a smaller project-based STEM campus feel, robotics lab, collaborative students, coding screens, and hands-on engineering projects.
- Right side: Georgia Tech, represented by a larger urban research university feel, advanced technology buildings, robotics, computing labs, and a busy campus environment. The mood should be thoughtful, balanced, and supportive, not biased toward either college. Make it look like a college decision comparison, not an advertisement.

Expected Answer: A side-by-side comparison between the two selected colleges, highlighting each of their good sides, must knows , and information/oportunities related to each college .

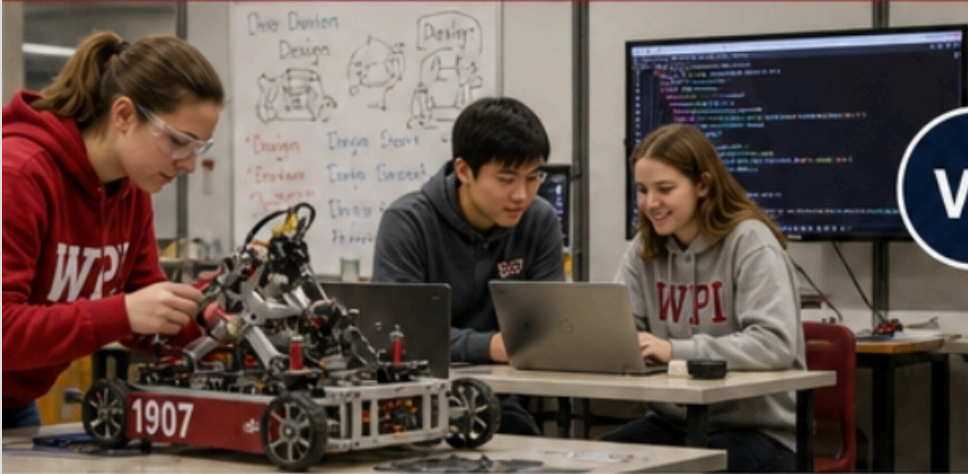
COLLEGE DECISION COMPARISON

Computer Science • Robotics • Engineering • Scholarships • Affordability • Overall Fit

WPI

WORCESTER
POLYTECHNIC
INSTITUTE

GEORGIA
INSTITUTE
OF TECHNOLOGY



VS.

 Strong CS fundamentals with hands-on, project-based curriculum. Opportunities in AI, systems, robotics, and more.	 COMPUTER SCIENCE	 Top-ranked CS program with deep specializations in AI, ML, systems, and theory. Strong industry connections.
 Hands-on robotics from day one. Small teams, direct faculty mentorship, and strong maker culture.	 ROBOTICS & ENGINEERING	 State-of-the-art labs and research centers. Large-scale projects and nationally recognized teams.
 Collaborative, tight-knit community. ~4,700 undergrads. Active learning and close faculty interaction.	 LEARNING ENVIRONMENT	 Large, energetic campus with diverse perspectives. Many opportunities to get involved and lead.
 Merit-based scholarships available. Competitive awards for high-achieving students.	 SCHOLARSHIPS & AID	 Strong merit-based scholarships. Programs like HOPE and other need-based aid available.
 Sticker price \$60K+ Generous aid and merit can make WPI very affordable.	 AFFORDABILITY (ESTIMATED)	 Sticker price \$33K+ (in-state) Out-of-state higher; scholarships and aid can significantly reduce cost.
 Worcester, MA. Smaller city with a close campus. Easy access to Boston (tech, internships, culture).	 LOCATION & CAMPUS	 Atlanta, GA. Major urban center with thriving tech industry and internship opportunities.
 OVERALL FIT: WPI Great for students who want a close-knit, project-driven community with hands-on learning and strong mentorship.		 OVERALL FIT: GEORGIA TECH Great for students who want a large, dynamic research university with extensive resources and a strong global network.



Both are excellent choices. Think about your preferred learning environment, location, and financial package to decide which is the best fit for YOU.

Live Demo / Code Overview

What we learned

Thanks!

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